PROJECT REPORT

GLOBAL OLYMPIC SPORTS PERFORMANCE ANALYSIS

Using Tableau

Submitted by
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Team ID
To be provided

Project domain
Data Analytics / Business Intelligence

Primary tool
Tableau

Academic Project Documentation

Report prepared using the project details and evidence available as of 2 November 2026.

DOCUMENTATION NOTE

This report consolidates the project information shared so far, including ideation, requirements, design, planning, development, Tableau dashboards/story structure and performance-testing status. It does not invent screenshots, measured timings, public links, GitHub repositories, Snowflake execution results or numerical findings that were not supplied.

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1. INTRODUCTION

1.1 Project Overview

Global Olympic Sports Performance Analysis is a data-analytics project designed to explore historical Olympic participation and medal information. The project uses Tableau to turn athlete-level records into interactive visualizations and dashboards that help users examine medal performance across countries, sports, years and seasons.

The project is organized around three stakeholder scenarios: Priya, a National Olympic Committee Performance Strategist; Carlos, a Sports Investment Analyst; and William, an Olympic Data Storyteller. Each scenario represents a different way of using Olympic data to answer practical questions.

The available athlete-events dataset contains 271,116 records and 15 columns. Its known coverage is 1896–2016 and includes both Summer and Winter Olympic Games. Key fields include ID, Name, Sex, Age, Height, Weight, Team, NOC, Games, Year, Season, City, Sport, Event and Medal.

1.2 Purpose

• Convert historical Olympic data into understandable charts and dashboards.

• Compare medal performance by country/NOC, sport, year and season.

• Explore athlete participation and average medals per athlete.

• Present insights through a Tableau Story for different stakeholder scenarios.

• Provide a foundation for future analysis using verified regional, host-country and sports-investment data.

Project objectives

Objective	Expected contribution
Explore Olympic history	Show changes in medal outcomes over time and between Summer/Winter Games.
Compare NOCs	Identify countries/NOCs with stronger medal performance in selected periods.
Compare sports	Study medal distribution and average medals per athlete by sport.
Support decisions	Provide visual evidence for performance discussion; investment conclusions require additional data.
Communicate findings	Use dashboards and four story points to explain the analysis.

2. IDEATION PHASE

2.1 Problem Statement

Olympic results are distributed across many years, sports, events and participating countries. Reviewing raw athlete-level records manually makes it difficult to compare performance, identify trends and communicate findings clearly. The project addresses this problem by organizing the data into interactive Tableau visualizations that support exploration and comparison.

2.2 Empathy Map Canvas

Perspective	Stakeholder needs and observations
Says	“I need a clear way to compare Olympic performance across countries, sports and years.”
Thinks	Which sports and countries perform strongly? Are medal patterns changing over time?
Does	Uses filters, compares charts, reviews trends and explores dashboard story points.
Feels	Needs confidence that the data is accurate, understandable and not misleading.
Pain points	Large raw dataset; repeated team/event medal rows; missing region/investment fields; complex comparisons.
Gains	Faster visual comparison, clearer trends and a structured presentation of historical data.

2.3 Brainstorming

Ideas considered for the project include country-wise medal ranking, average medals per athlete by sport, medal distribution by sport, medal trends by year and season, host-nation comparison, regional medal distribution, and sports-investment efficiency. The ideas were grouped into three stakeholder scenarios and arranged into three dashboards.

Idea prioritization

Priority	Idea	Reason
High	Country/NOC ranking; medal distribution; year/season trends	Can be explored using available dataset fields.
High	Average medals per athlete by sport	Can be derived from athlete ID and medal information, subject to validation.
Medium	Host-nation comparison	Requires a reliable host-country/year lookup and careful medal-count logic.
Future	Regional medal efficiency	Requires a verified NOC-to-region mapping.
Future	Investment vs medal performance	Requires comparable sports investment data that is not in the base CSV.

3. REQUIREMENT ANALYSIS

3.1 Customer Journey Map

Stage	User activity	Need / expected experience
Discover	Opens the Olympic analysis project.	Understand the project purpose and dataset coverage.
Explore	Selects a dashboard and reviews charts.	Clear titles, legends, labels and readable layout.
Filter	Filters by year, season, NOC, sport or medal.	Relevant views should update consistently.
Compare	Compares countries, sports and periods.	Comparable metrics and transparent calculations.
Understand	Moves through Tableau Story points.	A logical explanation for each stakeholder scenario.
Decide / Share	Uses the charts in discussion or presentation.	Accurate caveats and easy-to-read results.

3.2 Solution Requirement

Functional requirements

• Load and analyze the historical Olympic athlete-events dataset.

• Display medal performance by country/NOC, sport, year and season.

• Provide interactive filters for Year, Season, NOC, Sport and Medal.

• Show three dashboards and a four-point Tableau Story.

• Use calculated fields for medal flags and per-athlete measures where appropriate.

Non-functional requirements

• Readable charts, consistent colors, clear labels and navigation.

• Reasonable dashboard responsiveness on the computer used for evaluation.

• Correct aggregation and transparent treatment of missing or repeated data.

• Avoid unsupported conclusions when supplementary data is unavailable.

3.3 Data Flow Diagram

Level 0 – Context flow

Input	Process	Output
Olympic athlete-events CSV	Import data into Tableau; inspect fields; prepare calculated measures; build visualizations.	Interactive worksheets, dashboards and Tableau Story.

Level 1 – Main processing steps

1. Obtain the athlete-level CSV → 2. Check field names, data types and coverage → 3. Create medal-related calculations → 4. Build worksheets → 5. Arrange worksheets into dashboards → 6. Add filters and story points → 7. Validate counts and interpretations → 8. Present results.

Data-quality caution: athlete-level rows can repeat a team medal for multiple athletes. Medal totals should therefore use a carefully validated distinct event/medal key rather than blindly counting all non-null Medal rows.

3.4 Technology Stack

Technology / component	Role in project	Status
Tableau	Data visualization, dashboards, filters and story presentation.	Primary platform used for project work.
CSV dataset	Historical athlete/event/medal source data.	Available; 271,116 rows and 15 columns.
Calculated fields	Create medal flag and per-athlete measures.	Discussed/used in Tableau workflow; validate final formulas.
Snowflake	Possible cloud data warehouse for future data preparation.	No implementation evidence supplied.
Streamlit / web app	Possible future interactive deployment.	Not confirmed as implemented.

4. PROJECT DESIGN

4.1 Problem-Solution Fit

The problem is the difficulty of interpreting a large historical dataset using raw rows and separate records. The proposed solution is a Tableau-based visual analysis environment. It fits the need because dashboards allow users to filter, compare and interpret results without manually scanning the dataset. The solution is strongest for questions supported by the available fields; region, host and investment analyses need supplementary data.

4.2 Proposed Solution

The solution organizes the analysis into three dashboards and a four-point story. The dashboard plan is as follows:

Dashboard	Visualization	Purpose / stakeholder
1. National Olympic Performance	1. Average medals per athlete by sport	Compare sport-level athlete medal averages; Priya.
1. National Olympic Performance	2. Country/NOC medal ranking	Compare medal performance across NOCs; Priya.
1. National Olympic Performance	3. Medal distribution by sport (treemap)	Show how medals are distributed across sports; Priya.
2. Sports Investment & Regional Efficiency	4. Investment vs medal performance	Explore investment-performance relationship; Carlos. Requires investment data.
2. Sports Investment & Regional Efficiency	5. Regional medal efficiency	Compare regions; requires verified NOC-region mapping.

Dashboard	Visualization	Purpose / stakeholder
2. Sports Investment & Regional Efficiency	6. Country-wise medal efficiency	Compare country efficiency; define denominator and source data clearly.
3. Olympic Trends & Storytelling	7. Medal trends by year/season	Show historical change across Olympic editions; William.
3. Olympic Trends & Storytelling	8. Host nation comparison	Compare host performance; requires host lookup data.
3. Olympic Trends & Storytelling	9. Regional medal distribution by sport	Compare sport patterns by region; requires region mapping.

Suggested dashboard filters: Year, Season, NOC, Sport and Medal. Add Region only after the mapping data has been sourced and validated. A light theme with white/light-blue backgrounds, navy headings, sky-blue accents and gold/silver/bronze medal colors is the preferred design direction.

4.3 Solution Architecture

Layer	Description
Data source	Historical Olympic athlete-events CSV (271,116 rows; 15 fields; 1896–2016).
Data preparation	Review schema, data types, nulls, duplicate records and medal aggregation. Supplementary lookup tables are needed for region, host and investment questions.
Analytics	Tableau calculated fields, dimensions, measures, filters and worksheet-level analysis.
Presentation	Three Tableau dashboards and one Tableau Story with four story points.
User	Stakeholders explore, filter, compare and interpret the visual results.

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

The project work is divided into phases so that the problem, requirements, design, development and testing are completed in a logical order. The schedule below is a phase-based plan; exact calendar dates and individual task durations were not provided.

Phase	Main activities	Deliverable / status
1. Ideation	Identify problem, stakeholders, brainstorm ideas and prioritize visual questions.	Ideation templates prepared earlier.
2. Requirement analysis	Define user journey, functional/non-functional requirements, data flow and technology stack.	Requirement-analysis templates prepared earlier.
3. Project design	Define solution fit, dashboards, architecture and story structure.	Project-design templates prepared earlier.
4. Planning	Organize tasks, dependencies, risks and expected deliverables.	Planning phase was discussed; final source template completion not confirmed.
5. Development	Prepare data, create worksheets, dashboards, filters and story points.	Development report compiled; some supplementary analyses remain dependent on additional data.
6. Testing	Check data quality, calculations, filters, dashboards and story navigation.	Test plan documented; actual timings/screenshots not supplied.
7. Documentation	Prepare final report, appendices and evidence.	This report.

Dependencies and risks

• Correct medal aggregation is essential because team/event medals may be repeated on athlete rows.

• Region, host and investment dashboards depend on supplementary datasets and reliable join keys.

• The provided CSV ends in 2016; later Olympic editions must not be claimed without additional data.

• Dashboard performance depends on the computer, Tableau version, data connection and workbook design.

• Final screenshots, public links and measured test values must be captured from the actual project environment.

6. FUNCTIONAL AND PERFORMANCE TESTING

6.1 Performance Testing

Testing is intended to confirm that the data is interpreted correctly, filters behave as expected, calculations are valid, and dashboards/story points display correctly. No measured execution times or system specifications were provided, so those values are left pending instead of being fabricated.

Test area	Test / expected result	Evidence status
Data rendering	Dataset loads and displays the expected fields and record volume.	Dataset facts known; actual load time pending.
Preprocessing	Check nulls, duplicates, data types and medal/event aggregation.	Executed test counts not supplied.
Filters	Year, Season, NOC, Sport and Medal filters update relevant views; clear restores full data.	Test execution evidence pending.
Calculated fields	Medal Flag and Medals per Athlete return valid results on a checked sample.	Formulas discussed; final sample validation pending.
Dashboard rendering	All three dashboards open without blank/error views and remain readable.	Three-dashboard structure reported; measured timings/screenshots pending.
Story navigation	All four story points navigate and show the correct dashboard/story content.	Four-point structure reported; evidence pending.
Snowflake	Database, warehouse, SQL queries and query history are tested only if Snowflake is implemented.	Not evidenced; Not applicable at present.

Known Tableau calculations

Medal Flag – a binary measure to identify rows with a medal value:

```
IF NOT ISNULL([Medal]) AND [Medal] <> 'NA' THEN 1 ELSE 0 END
```

Medals per Athlete – a simple ratio using distinct athlete IDs:

```
SUM([Medal Flag]) / COUNTD([ID])
```

These formulas are starting points. The exact final workbook calculations and medal-counting logic should be checked against a manually verified sample, especially for team events and ties.

Performance metrics to capture during final live testing

• Time to open each dashboard and the full Tableau Story.

• Time for each filter to update the views.

• Dataset file size / extract size and computer specifications.

• Number of records before and after any cleaning steps.

• Screenshots of successful dashboard views and filter interactions.

7. RESULTS

7.1 Output Screenshots

The project plan includes three dashboards and a four-point Tableau Story. The user has previously reported completing the dashboards/story structure, but actual image files or workbook screenshots were not included with the documentation materials available for this report. Therefore, the following evidence slots identify what should be inserted from the live Tableau workbook.

Evidence slot	Screenshot to insert	Caption
Figure 1	[Insert screenshot of Dashboard 1]	National Olympic Performance: athlete averages, NOC ranking and sport medal distribution.
Figure 2	[Insert screenshot of Dashboard 2]	Sports Investment and Regional Efficiency; include only views supported by verified supplementary data.
Figure 3	[Insert screenshot of Dashboard 3]	Olympic Trends and Storytelling: year/season trends, host and regional views where data is available.
Figure 4	[Insert screenshot of Tableau Story]	Four story points: Project Overview, Scenario 1, Scenario 2 and Scenario 3.
Figure 5	[Insert screenshot of filters in use]	Demonstrate Year, Season, NOC, Sport and Medal filters.

Expected analytical outputs

The project is designed to let viewers identify leading NOCs in a selected period, compare medal distribution across sports, examine medal trends over Olympic editions and inspect average medal values relative to athlete participation. Specific rankings, trends and numerical conclusions are intentionally not stated here because the final workbook output values were not supplied for independent verification.

Dataset limitations affecting results

• The supplied dataset covers 1896–2016, not 2020–2026.

• The base dataset does not contain a Region field, host-country lookup or sports-investment values.

• Repeated athlete rows in team events can inflate raw medal counts if counted incorrectly.

• Any regional, host or investment finding must cite the supplementary dataset and its join method.

8. ADVANTAGES & DISADVANTAGES

Advantages

• Converts a large historical dataset into visual summaries that are easier to understand.

• Allows interactive filtering by key Olympic dimensions.

• Supports comparisons across countries/NOCs, sports, years and seasons.

• Uses a story-based structure to communicate analysis to different stakeholder types.

• Can be extended with verified supplementary datasets.

Disadvantages / limitations

• The base CSV does not contain investment or region fields, limiting some planned views.

• Host-nation analysis requires a separate verified host lookup.

• Historical coverage stops at 2016 in the supplied file.

• Medal aggregation requires care to avoid duplicate counting in team events.

• Performance metrics and screenshot evidence still need to be captured from the actual workbook.

9. CONCLUSION

Global Olympic Sports Performance Analysis demonstrates how Tableau can be used to explore historical Olympic athlete and medal data through interactive dashboards and a structured Story. The project connects data analysis with three stakeholder scenarios and organizes the work into nine planned visualizations across three dashboards. The available data supports several historical comparisons, while regional, host and investment-focused conclusions require additional verified data. The report documents the project design and current evidence honestly and identifies the remaining validation and submission tasks.

10. FUTURE SCOPE

• Add Olympic data for editions after 2016 using a reliable source and consistent schema.

• Integrate an NOC-to-region mapping to enable regional comparisons.

• Add an authoritative host-city/host-country lookup for host performance analysis.

• Add comparable sports funding/investment data with clear year, country and sport definitions.

• Improve medal aggregation using a validated event-level medal key and test for ties/team events.

• Publish the Tableau workbook or a companion web application if permitted by the course/institution.

• Add automated data-quality checks and record performance metrics for each dashboard.

• Refresh the dashboard as new Olympic datasets become available.

11. APPENDIX

A. Source code / calculated fields

The project is primarily a Tableau visualization project. No standalone application source code or Git repository was supplied. The following Tableau calculated-field examples were discussed during development and should be checked against the final workbook before submission.

1. Medal Flag

```
IF NOT ISNULL([Medal]) AND [Medal] <> 'NA' THEN 1 ELSE 0 END
```

2. Medals per Athlete

```
SUM([Medal Flag]) / COUNTD([ID])
```

B. Dataset link

Dataset file used: athlete-level Olympic events CSV (the supplied project dataset). A public download URL was not included in the materials available for this report. Add the exact dataset source URL and access date here if required by the instructor.

Dataset fields: ID, Name, Sex, Age, Height, Weight, Team, NOC, Games, Year, Season, City, Sport, Event and Medal. Known coverage: 1896–2016; 271,116 records; 15 columns.

C. GitHub and project demo link

GitHub repository: Not provided.
Tableau Public / project demo link: Not provided.
Team ID: To be provided.

D. Additional validation notes

• Do not describe Snowflake or Streamlit as implemented unless a working instance and evidence are available.

• Do not insert fabricated screenshots. Capture images directly from the final Tableau workbook.

• Record the actual Tableau version, operating system and computer configuration if performance testing is assessed.

• Attach the original dataset citation and any supplementary data source citations in the final submission.

E. Project completion checklist

Item	Current documentation status
Ideation, requirements and design	Previously prepared project-phase documents are available in the project workflow.
Planning phase	Final planning template completion not confirmed.
Development documentation	Development phase report compiled with pending dependencies identified.
Performance testing	Testing report prepared; live timings/screenshots remain pending.
Final project documentation	This consolidated report.
Screenshots, public links and Team ID	Must be added by project owner where available.

End of Report